AgroFlow showcase v4: voice-over script

INF3003W Team 2 · video length 4:43.4 · 1920×1080, 25 fps · read at about 140 words a minute

Times are minutes:seconds from the start of the video. **In** is when your first word starts; **Out by** is when you must be finished (the next line starts, or the scene changes). Record one WAV per line, 48 kHz, quiet room, with 1 second of silence at the start, named as shown.

Speaker	Lines
Leruo	1 problem · 6 field kit · 9 execution · 10 weather change · 13 testing · close (b)
Olwethu	2 enquiry · 3 call-back · 11 reports · 12 admin view · close (d)
Philisile	4 onboarding · 5 Telegram and first sign-in · close (c)
Lubaelihle	7 zone setup in the app · 8 daily use · close (a)

1. The problem · Leruo

File: **s01_leruo.wav** · In: **0:02.6** · Out by: **0:15.8** (slot 13.2 s) · 29 words ≈ 12.4 s at 140 wpm

On many small farms, watering is guesswork. Too much wastes water and electricity; too little stresses the crop. AgroFlow helps a farmer decide when to irrigate, and how much.

2. The enquiry · Olwethu

File: **s02_olwethu.wav** · In: **0:16.3** · Out by: **0:34.7** (slot 18.4 s) · 41 words ≈ 17.6 s at 140 wpm

Meet Thabo, one of our personas: a tomato farmer near Mafikeng, with one field split into two irrigation zones. He asks for a call back on the AgroFlow website. If he mistypes his phone number, the form tells him straight away.

3. The call-back · Olwethu

File: **s03_olwethu.wav** · In: **0:35.2** · Out by: **0:53.1** (slot 17.9 s) · 40 words ≈ 17.1 s at 140 wpm

Naledi, our second persona, is a field officer. Telegram tells her a new enquiry has arrived. In the Enquiries tab she taps "I'll handle this", so nobody else calls him too. After the call, she logs it: Thabo wants AgroFlow.

4. Onboarding · Philisile

File: **s04_philisile.wav** · In: **0:53.6** · Out by: **1:12.4** (slot 18.8 s) · 42 words ≈ 18.0 s at 140 wpm

She starts onboarding straight from the enquiry, so Thabo's details are already filled in. She adds his farm and both zones, with crop, soil and size. The enquiry then shows he became a farmer, and the activity log records who did it.

5. Telegram and first sign-in · Philisile

File: **s05_philisile.wav** · In: **1:12.9** · Out by: **1:32.6** (slot 19.7 s) · 44 words ≈ 18.9 s at 140 wpm

To hand over, Naledi shows a QR code. Thabo scans it and presses Start in the AgroFlow Telegram bot, which sends his sign-in email, a temporary password and the app link. When he first signs in, the app makes him choose his own password.

6. The field kit · Leruo

File: **s06_leruo.wav** · In: **1:33.1** · Out by: **1:58.8** (slot 25.7 s) · 58 words ≈ 24.9 s at 140 wpm

Next, the field kit: an ESP32 board, a Soil Moisture Sensor, a temperature and humidity sensor, and a relay for the pump. Naledi registers the kit to Zone 1 and gets a one-time code. On site, the installer joins the kit's setup Wi-Fi, enters the farm Wi-Fi and the code, and the kit links itself to the zone.

7. Zone setup in the app · Lubaelihle

File: **s07_luba.wav** · In: **1:59.3** · Out by: **2:23.2** (slot 23.9 s) · 54 words ≈ 23.1 s at 140 wpm

In the app, Thabo finishes each zone's setup. For the flow rate, he times how long his tap takes to fill a 500 millilitre bottle. Then he teaches the sensor his soil, with a wet and a dry reading on different days. With each step, the advice gets surer: from medium to high confidence.

8. Daily use · Lubaelihle

File: **s08_luba.wav** · In: **2:23.7** · Out by: **2:45.5** (slot 21.8 s) · 49 words ≈ 21.0 s at 140 wpm

Every day, Home shows one action per zone. Today, Zone 1 needs about 66 minutes of water, or 996 litres, with high confidence. "Why am I seeing this?" shows the reasons: the weather, the crop's stage and the water left in the soil. The week ahead plans each day.

9. Execution · Leruo

File: **s09_leruo.wav** · In: **2:46.0** · Out by: **3:10.4** (slot 24.4 s) · 55 words ≈ 23.6 s at 140 wpm

AgroFlow recommends; Thabo decides. He starts a run from the app. The command reaches the kit over MQTT, the relay switches the pump on, and the kit reports the run back. For safety, the kit never runs the pump for more than five minutes at a time. Our website demo shows this as a visualisation.

10. The weather changes · Leruo

File: **s10_leruo.wav** · In: **3:10.9** · Out by: **3:31.8** (slot 20.9 s) · 47 words ≈ 20.1 s at 140 wpm

Then the forecast changes: 24 millimetres of rain tomorrow. AgroFlow works the plan out again, and Zone 2 no longer needs water on Monday. If rain covers the crop's need, it says wait; if it only helps, it says water less. Thabo also gets a Telegram alert.

11. Reports · Olwethu

File: **s11_olwethu.wav** · In: **3:32.3** · Out by: **3:49.8** (slot 17.5 s) · 39 words ≈ 16.7 s at 140 wpm

Reports start with what to do next. Thabo makes his as a PDF in the app, and automatic ones arrive on Telegram. Naledi can preview a farm's report, download PDF or Excel, and schedule daily, weekly or monthly reports.

12. Naledi's admin view · Olwethu

File: **s12_olwethu.wav** · In: **3:50.3** · Out by: **4:05.7** (slot 15.4 s) · 34 words ≈ 14.6 s at 140 wpm

For Naledi, Home shows what needs attention across every farm: new requests, quiet kits and farmers without Telegram. Every action is logged, so a small team can look after many farms without losing track.

13. Testing and validation · Leruo

File: **s13_leruo.wav** · In: **4:06.2** · Out by: **4:27.1** (slot 20.9 s) · 44 words ≈ 18.9 s at 140 wpm

We tested as we built. Forms reject bad input, like a wrong phone number or a run over 60 minutes. In our local test runs on 9 and 10 October, all 1,301 backend tests, all 112 admin tests and all 64 mobile tests passed.

14. Close (a) · Lubaelihle

File: **s14a_luba.wav** · In: **4:27.7** · Out by: **4:31.0** (slot 3.3 s) · 5 words ≈ 2.1 s at 140 wpm

AgroFlow recommends. The farmer decides.

14. Close (b) · Leruo

File: **s14b_leruo.wav** · In: **4:31.0** · Out by: **4:34.5** (slot 3.4 s) · 7 words ≈ 3.0 s at 140 wpm

Every number traces back to its source.

14. Close (c) · Philisile

File: **s14c_philisile.wav** · In: **4:34.5** · Out by: **4:39.0** (slot 4.5 s) · 6 words ≈ 2.6 s at 140 wpm

Built by Team 2 for INF3003W.

14. Close (d) · Olwethu

File: **s14d_olwethu.wav** · In: **4:39.0** · Out by: **4:43.4** (slot 4.4 s) · 4 words ≈ 1.7 s at 140 wpm

Thank you for watching.